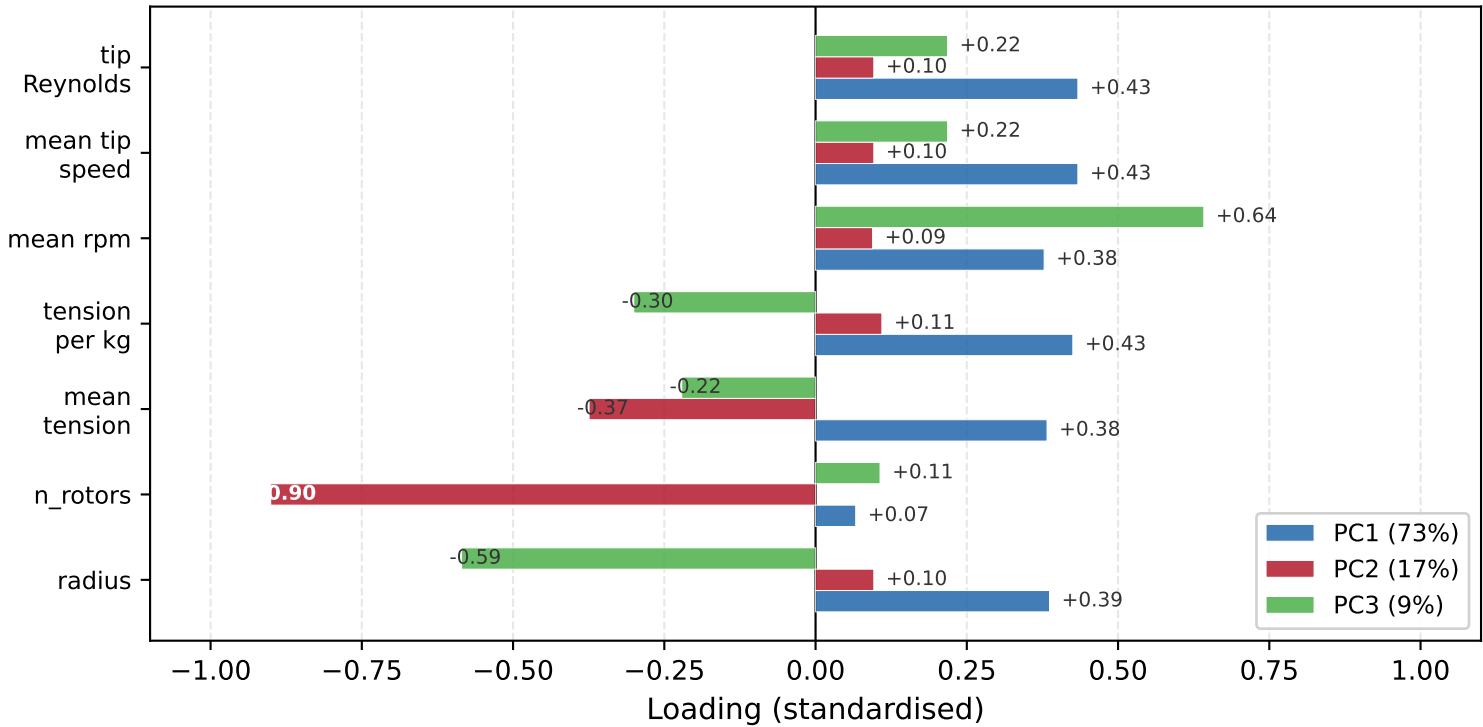


PCA Loadings: What Drives Each Principal Component?



IMPLICATIONS: PC1 loads heavily on radius, tip speed, Reynolds, RPM, and tension per kg — all positive, all collinear. These 5 variables measure the same underlying physical dimension: rotor scale. PC2 is dominated by n_rotors (+0.99) — adding rotors is an independent design choice from physical scale. PC3 loads on elevation angle. The key takeaway for instrumentation: you don't need 7 sensors. One radius measurement and one RPM sensor capture >80% of the design variance. Data: 96 configurations from BEM v2.0 sweep.